UNIT- V Integrative Engineering Design Solutions

Identifying and resolving issues with working in diverse teams, Modularising, prototype building by different engineering disciplines within the team, validated learning with accessible metrics Capstone Project (Interdisciplinary)
Applying Design Thinking Principles and Methods for Ideation and Prototyping, Testing Solution, Refining Solution, and Taking the Solution to the Users

🔷 1. Introduction

Integrative Engineering Design focuses on **combining multiple engineering disciplines** (mechanical, electrical, software, etc.) to solve complex real-world problems.
It emphasizes:

- Collaboration
- Innovation
- User-centered design
- Practical implementation

🔷 2. Working in Diverse Teams

✅ Definition

A **diverse team** includes members from different:

- Engineering branches
- Skill sets
- Cultural backgrounds

⚠️ Common Issues

- Communication gaps
- Conflicting ideas
- Role ambiguity
- Lack of coordination

✅ Solutions

- Clear role allocation
- Regular team meetings
- Use of collaboration tools
- Leadership & conflict resolution

🔷 3. Modularization in Design

✅ Definition

Breaking a complex system into **smaller independent modules**.

✅ Advantages

- Easy development
- Parallel work by teams
- Simplified testing
- Easy maintenance

📌 Example

Smart irrigation system:

- Sensor module
- Control unit
- Water pump system

🔷 4. Prototype Building

✅ Definition

Creating a **preliminary model** of a product to test ideas.

🔧 Types

- Low-fidelity (sketch, paper model)
- High-fidelity (working model, simulation)

✅ Role of Different Disciplines

- Mechanical → Structure design
- Electrical → Circuit design
- Software → Control system

🎯 Purpose

- Validate concept
- Identify design flaws early

> **"Low-fidelity prototypes are used in early stages for idea generation, while high-fidelity prototypes are used in later stages for testing and validation."**

🔷 1. Low-Fidelity Prototype

✅ Definition

A **low-fidelity prototype** is a **simple, rough, and inexpensive model** used to quickly represent an idea or concept.

🔧 Characteristics

- Basic design (not detailed)
- Low cost and quick to create
- No or minimal functionality
- Easy to modify

🛠️ Examples

- Hand-drawn sketches
- Paper models
- Wireframes (basic UI layouts)
- Cardboard mock-ups

📌 Engineering Example

👉 **Smart Irrigation System**

- Draw a block diagram showing sensor → controller → pump
- Paper layout of system components

🎯 Purpose

- Quickly visualize ideas
- Get early feedback
- Identify basic design issues

🔷 2. High-Fidelity Prototype

✅ Definition

A **high-fidelity prototype** is a **detailed, functional model** that closely resembles the final product.

🔧 Characteristics

- Realistic design and functionality
- More expensive and time-consuming
- Includes working components
- Provides accurate testing results

🛠️ Examples

- Working electronic circuit
- CAD simulation models
- Fully functional app/software
- 3D printed models

📌 Engineering Example

👉 **Smart Irrigation System**

- Soil moisture sensor connected to Arduino
- Automatic water pump control
- Real-time operation in field

🎯 Purpose

- Test real performance
- Validate design accuracy
- Demonstrate final product

🔷 5. Validated Learning & Accessible Metrics

✅ Validated Learning

Learning based on **real data and testing**, not assumptions.

📊 Accessible Metrics

Measurable indicators such as:

- Efficiency
- Cost
- User satisfaction
- Performance accuracy

🎯 Importance

- Helps in decision making
- Improves product quality
- Ensures real-world applicability

Describe the complete process of a capstone project using Design Thinking methodology for developing a **Smart Irrigation System for farmers**. Explain all stages Empathize, Define, Ideate, Prototype, Test and Implementation.

🔷 1. Empathize (Understand Users)

Objective: Study user needs and problems

Process:

- Farmers are interviewed about irrigation challenges
- Observations reveal water wastage and lack of automation

Tools Used:

- Empathy Map
- Field Survey

Outcome:

Farmers need a **low-cost automated irrigation system.**

🔷 2. Define (Problem Statement)

Objective: Clearly define the problem

Problem Statement:

👉 *"Farmers need an efficient irrigation system to reduce water wastage and improve crop yield."*

Tools Used:

- Affinity Diagram
- 5 Whys Analysis

🔷 3. Ideate (Generate Solutions)

Objective: Develop multiple ideas

Ideas Generated:

- Soil moisture sensor-based system
- Mobile-controlled irrigation
- Timer-based irrigation

Tools Used:

- Brainstorming
- Mind Mapping

Final Idea Selected:

👉 **Automatic irrigation using soil moisture sensors**

4. Prototype (Build Model)

Objective: Create a working model

Prototype Developed:

- Soil moisture sensor
- Microcontroller (Arduino)
- Water pump system

Tools Used:

- Circuit Design
- CAD/Block Diagram

5. Test (Evaluate Solution)

Objective: Check performance

Testing Done:

- Tested in farm conditions
- Measured water usage and efficiency

Tools Used:

- User Feedback
- Performance Metrics

Result:

Water usage reduced significantly.

6. Refine (Improve Design)

Objective: Improve based on feedback

Improvements:

- Added mobile alerts
- Improved sensor accuracy
- Reduced system cost

7. Implement (Deploy Solution)

Objective: Deliver to users

Deployment:

- Installed in farms
- Provided user training

Outcome:

- Increased crop yield
- Reduced manual effort

8. Interdisciplinary Contribution

- **Electronics:** Sensor and circuit design
- **Mechanical:** Pump and pipeline system
- **Software:** Control logic and automation

Describe the complete process of a capstone project using Design Thinking methodology for developing a **Smart Waste Management System.**

🔷 Problem Chosen

👉 Inefficient waste collection leads to **overflowing bins, unhygienic conditions, and increased pollution.**

🔷 1. Empathize (Understand Users)

Objective: Understand the needs of users (municipal workers, residents).

Process:

- Interview sanitation workers and residents
- Observe waste collection issues (overflow, delays)

Tools Used:

- Empathy Map
- Surveys & Field Observation

Outcome:
Need for a **real-time waste monitoring system**.

🔷 2. Define (Problem Statement)

Objective: Clearly define the problem.

👉 **Problem Statement:**
"Urban areas need an efficient waste management system to monitor bin levels and optimize collection routes."

Tools Used:

- Affinity Diagram
- 5 Whys Analysis

🔷 3. Ideate (Generate Solutions)

Objective: Generate innovative ideas.

Ideas:

- Smart bins with sensors
- Mobile app for tracking waste levels
- GPS-based route optimization

Tools Used:

- Brainstorming
- Mind Mapping

Final Idea:

👉 **IoT-based Smart Waste Management System**

🔷 5. Test (Evaluate Solution)

Objective: Validate performance.

Testing:

- Check sensor accuracy
- Monitor data transmission
- Simulate waste collection scenarios

Tools Used:

- User feedback
- Performance metrics (accuracy, response time)

Result:

Efficient monitoring of bin levels.

🔷 4. Prototype (Build Model)

Objective: Develop a working model.

Prototype Components:

- Ultrasonic sensor (to detect bin level)
- Microcontroller (Arduino/ESP32)
- GSM/Wi-Fi module
- Mobile/web dashboard

Tools Used:

- Circuit Design
- CAD/Block Diagram

🔷 6. Refine (Improve System)

Objective: Improve based on feedback.

Improvements:

- Enhance sensor precision
- Add alert notifications
- Optimize power consumption

🔷 7. Implement (Deployment to Users)

Objective: Deliver final solution.

Deployment:

- Install smart bins in selected areas
- Provide dashboard to municipal authorities

Outcome:

- Reduced overflow
- Optimized collection routes
- Cleaner environment